

Retail Performance Intelligence Hub

Power BI Business Analytics & Executive Dashboard

TOOLS

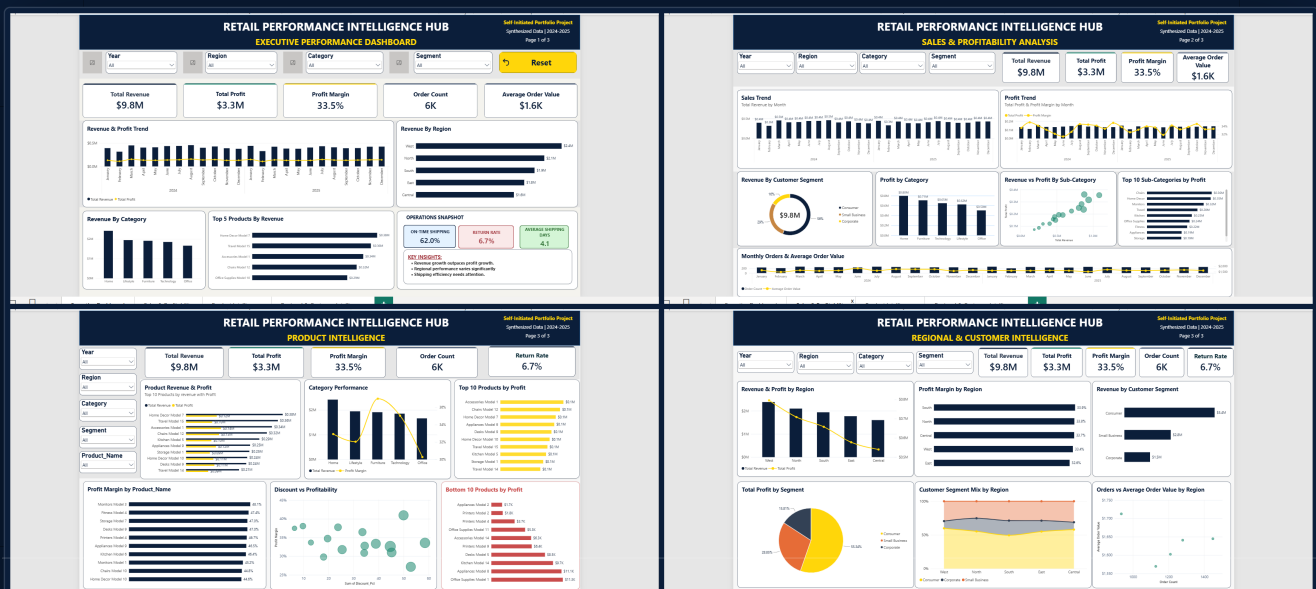
Power BI | Power Query |
DAX | Excel

FOCUS

Business Analytics & Executive
Reporting

DATA

Synthesized Retail Dataset,
2024–2025



CASE STUDY / 01

Project Overview

This project explores how a retail organization could use its transactional data to build a centralized business intelligence solution — one that lets decision-makers monitor revenue, profitability, product performance, regional performance, customer segments and operational metrics from a single, interactive dashboard.

This is a self-initiated portfolio project built using a synthesized dataset for demonstration purposes. It was designed to showcase how transactional retail data can be structured into a decision-support tool for business stakeholders, using Power BI, Power Query and DAX.

The dashboard is organized into four connected pages — an Executive Dashboard, a Sales & Profitability view, a Product Intelligence view, and a Regional & Customer Intelligence view — each built to answer a distinct set of business questions while sharing a common set of filters and KPI definitions.

ROLE Business Analytics Consultant	TOOLS Power BI, Power Query, DAX, Excel	DATASET Synthesized Retail Dataset
DATA PERIOD 2024 – 2025	ORDERS 6,000	CUSTOMERS 1,200
PRODUCTS 80	PROJECT TYPE Self-Initiated Portfolio Project	

CASE STUDY / 02

Business Problem & Objectives

In the simulated scenario, a retail business generates large volumes of transactional data but lacks a centralized management dashboard. Decision-makers have no quick way to understand sales performance, profitability, product performance, regional performance or customer-segment behavior without manually assembling reports from raw data.

KEY OBJECTIVES

- 1 Consolidate business performance metrics into a single, interactive dashboard.
- 2 Track revenue, profit and profitability trends over time.
- 3 Identify high- and low-performing products.
- 4 Compare performance across regions.
- 5 Understand how customer segments contribute to revenue and profit.
- 6 Analyze the relationship between discounting and profitability.
- 7 Provide a clear, executive-level view of overall business performance.

CASE STUDY / 03

Data & Analytics Approach

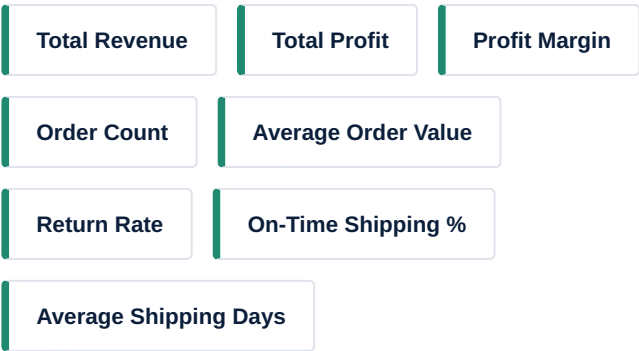
The dashboard was built by moving the synthesized retail dataset through a standard business intelligence workflow — from raw data to a decision-ready analytical product.



TECHNIQUES APPLIED

- Power Query for data preparation and transformation
- A relational data model connecting orders, products, customers and dates
- DAX measures for KPI calculation
- Interactive slicers for Year, Region, Category and Segment
- Cross-filtering data visualizations across each dashboard page
- Structured, page-level business performance analysis

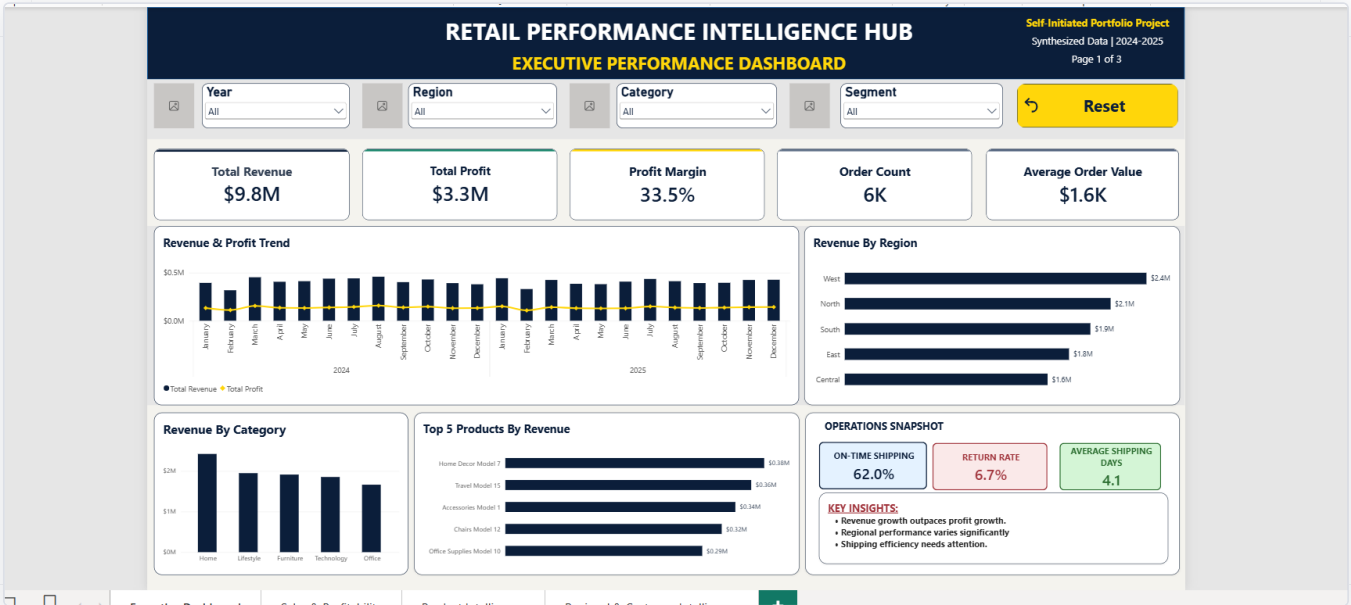
CORE KPI MEASURES



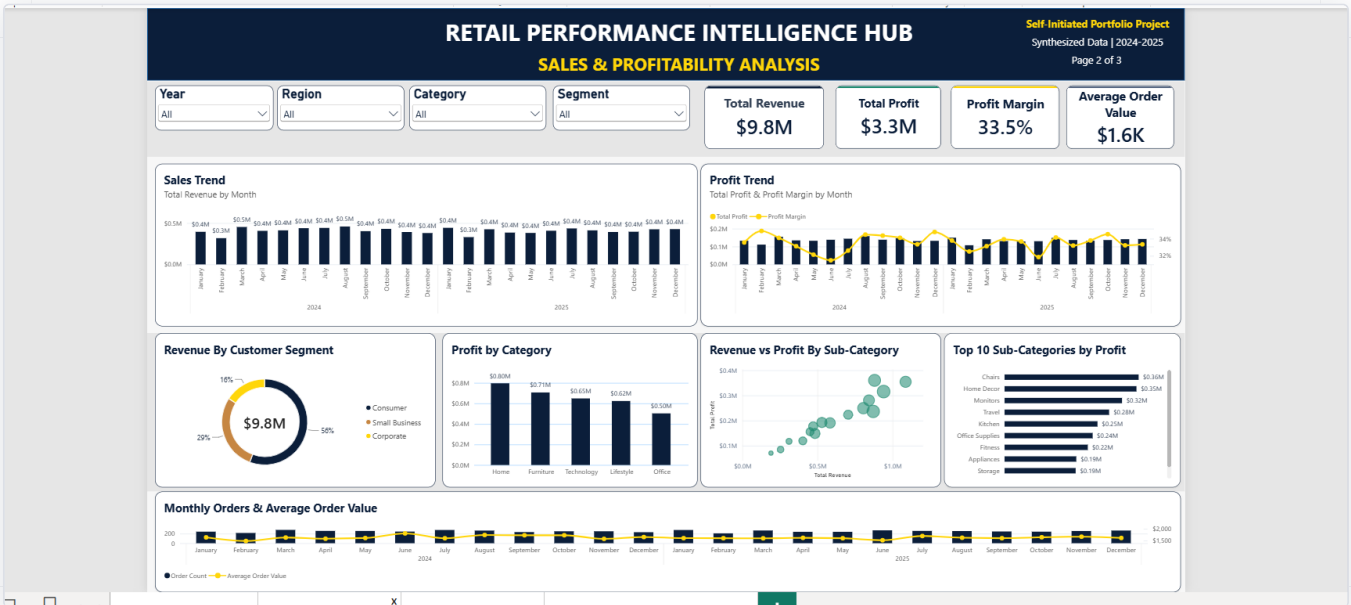
This project focuses on descriptive and diagnostic business analytics — understanding **what happened and why** across revenue, profitability, products, regions and customer segments. It does not include statistical modeling, machine learning, or forecasting.

CASE STUDY / 04

Dashboard Solution



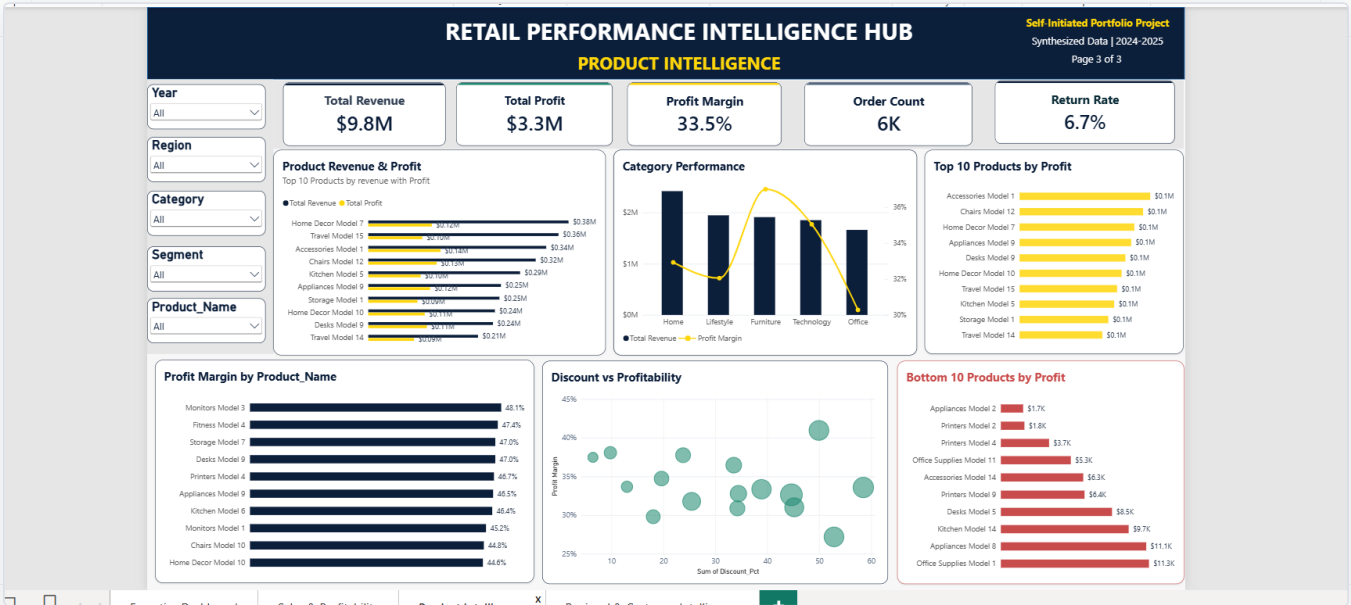
PAGE 1 **Executive Dashboard.** Provides a consolidated view of overall business performance and key operational indicators, with Year, Region, Category and Segment filters shared across the report.



PAGE 2 **Sales & Profitability.** Examines revenue and profitability drivers across time, categories, customer segments and sub-categories.

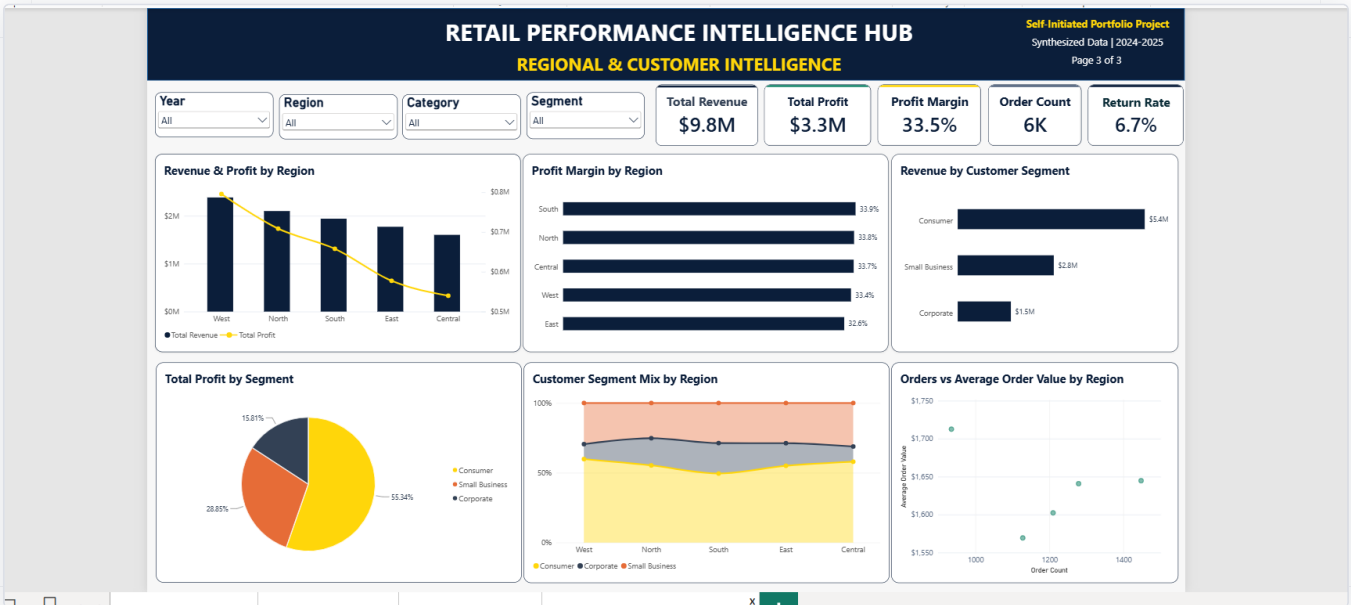
CASE STUDY / 04 (CONTINUED)

Dashboard Solution



PAGE 3

Product Intelligence. Provides product-level analysis of revenue, profit, margins, discounts and comparative product performance.



PAGE 4

Regional & Customer Intelligence. Compares regional performance and customer-segment contribution to understand where revenue and profitability are concentrated.

CASE STUDY / 05

Key Business Insights

INSIGHT 01 Revenue Growth Outpaces Profit Growth

The Executive Dashboard's own Key Insights panel flags that revenue growth is outpacing profit growth, alongside an overall Profit Margin of 33.5% on \$9.8M in total revenue and \$3.3M in total profit.

Business Interpretation: This pattern suggests that costs, discounting or product mix may be growing faster than revenue in parts of the business, and could be worth monitoring at the category or product level.

INSIGHT 02 Regional Performance Is Uneven

Revenue by Region ranges from \$2.4M in the West down to \$1.6M in Central, and Profit Margin by Region ranges narrowly from 33.9% (South) to 32.6% (East) — the lowest-margin region.

Business Interpretation: Revenue is more concentrated by region than profit margin is, which could indicate that regional teams are competing on similar profitability but at different scales of demand.

INSIGHT 03 Consumer Segment Drives Both Revenue and Profit

The Consumer segment contributes \$5.4M of the \$9.8M in total revenue and accounts for 55.34% of total profit, well ahead of Small Business (28.85%) and Corporate (15.81%).

Business Interpretation: Because Consumer contributes a similar share of both revenue and profit, this segment functions as the primary driver of the business overall, which may warrant closer monitoring than the smaller segments.

INSIGHT 04 A Small Set of Products Leads on Both Revenue and Profit

Home Decor Model 7, Travel Model 15, Accessories Model 1, Chairs Model 12 and Office Supplies Model 10 appear among the Top 5 Products by Revenue, while several of the same items also appear on the Top 10 Products by Profit list.

Business Interpretation: A small number of products appear to anchor both top-line revenue and profit contribution, which suggests they may be good candidates for inventory and marketing prioritization.

INSIGHT 05**Wide Spread Between Top and Bottom Product Profitability**

The Bottom 10 Products by Profit range from roughly \$1.7K to \$11.3K in profit, far below the Top 10 Products by Profit, and Profit Margin by Product ranges from about 44.6% at the top down toward the mid-20% range in the Discount vs Profitability view.

Business Interpretation: This spread indicates meaningful variation in product-level profitability that a category or merchandising review could help explain further.

INSIGHT 06**Discounting Shows a Loose Relationship With Margin**

The Discount vs Profitability scatter plot shows profit margins spanning roughly 25% to 45% across a discount range of about 10% to 60%, without a single clear linear trend.

Business Interpretation: The absence of a clean pattern suggests that discount level alone does not fully explain margin outcomes, and other factors (such as product or category) likely play a role.

INSIGHT 07**Operational Metrics Point to a Shipping Opportunity**

The Operations Snapshot shows On-Time Shipping at 62.0%, a Return Rate of 6.7%, and Average Shipping Days of 4.1 — and the dashboard's own key insights explicitly note that shipping efficiency needs attention.

Business Interpretation: On-time shipping performance stands out as the weakest operational metric on the dashboard, making it a natural starting point for further investigation into fulfillment and logistics.

CASE STUDY / 06

Skills Demonstrated

BUSINESS ANALYTICS

- KPI analysis and definition
- Revenue & profitability analysis
- Product performance analysis
- Customer segment analysis
- Regional performance analysis
- Business performance reporting

POWER BI

- Multi-page dashboard development
- Interactive slicers (Year, Region, Category, Segment)
- Data visualization design
- KPI cards and conditional formatting
- Cross-filtering between visuals

DATA

- Power Query data preparation
- Relational data modeling
- DAX measure creation
- Data cleaning and transformation

BUSINESS ANALYSIS

- Translating business questions into KPIs
- Structuring analytical dashboard views
- Identifying performance drivers
- Executive-oriented reporting

CASE STUDY / 07

Project Takeaways

"I can take business data, identify the questions that matter, structure relevant KPIs, build an interactive analytical solution, and communicate findings through a decision-oriented dashboard."

WHAT I BUILT

An interactive, 4-page Power BI business intelligence dashboard covering executive performance, sales & profitability, product intelligence, and regional & customer intelligence.

WHAT IT DEMONSTRATES

Business Analytics + Data Visualization + Business Analysis + Executive Reporting, built on a relational data model with DAX-driven KPIs.

Power BI

Power Query

DAX

Excel

PROJECT DISCLAIMER

This is a self-initiated portfolio project created using a synthesized dataset for demonstration and learning purposes. It does not represent client work, confidential business information, or actual company performance.